

Wearable Sensor-Based Mental Health Prediction in College Students Using Lightweight Machine Learning Models

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Abstract— Mental health issues such as stress and anxiety are increasingly prevalent among university students, negatively affecting academic performance and overall well-being. This paper presents a machine learning framework to predict daily stress levels from physiological and behavioral data collected via Fitbit Inspire 3 wearable devices. Using the SSAQS dataset comprising 35 undergraduate students monitored over one full academic semester, five models are implemented and compared: Random Forest, LightGBM, 1D Convolutional Neural Network, Long Short-Term Memory, and Fuzzy Logic. Input features include heart rate variability, blood oxygen saturation, daily step count, physical activity level, sleep quality, and Fitbit built-in stress score. All models are evaluated under Leave-One-Student-Out cross-validation to simulate deployment on unseen individuals. LightGBM achieves the highest accuracy of 0.3552 and weighted F1-score of 0.3532, while Fuzzy Logic achieves the lowest mean absolute error of 0.7400. Feature importance analysis identifies physical activity as the dominant predictor of self-reported stress, followed by blood oxygen saturation and heart rate variability.

Keywords—stress detection, wearable sensors, heart rate variability, machine learning, LOSO cross-validation, Fitbit, university students, fuzzy logic

I. INTRODUCTION

Mental health among university students has emerged as a critical public health concern worldwide. Chronic academic stress is associated with reduced cognitive performance, disrupted sleep, and increased risk of anxiety disorders and depression [1]. Traditional stress assessment methods rely predominantly on self-report questionnaires, which are administered infrequently, are subjective, and impose a burden on participants. Continuous, passive monitoring using consumer wearable devices offers a promising alternative that can capture physiological stress indicators throughout a student's natural daily routine.

The Fitbit Inspire 3 wearable device records a rich set of physiological and behavioral signals including heart rate variability (HRV), blood oxygen saturation (SpO2), physical activity, and sleep quality. HRV, which quantifies the variation in time between successive heartbeats, is widely recognized as a reliable non-invasive indicator of autonomic nervous system activity and psychological stress state [2]. When combined with complementary signals, these wearable-derived features provide a comprehensive representation of daily physiological load.

This paper makes the following contributions: (1) we apply five machine learning models to the SSAQS dataset [1], a longitudinal real-world dataset of 35 university students collected over one full academic semester; (2) we evaluate all

models under LOSO cross-validation to ensure generalization to unseen individuals; (3) we identify the most important sensor features for stress prediction using Random Forest feature importance analysis; and (4) we provide a comprehensive comparison of classical machine learning, deep learning, and rule-based approaches for this task.

II. RELATED WORK

Prior research on automated stress detection has primarily relied on physiological signals collected in controlled laboratory settings using research-grade equipment. Hellhammer et al. established cortisol and HRV as reliable biomarkers of acute psychological stress [3]. More recent studies have explored the use of consumer-grade wearables for stress monitoring in naturalistic settings, motivated by their low cost, unobtrusiveness, and widespread adoption.

Garcia-Ceja et al. demonstrated that combining heterogeneous wearable sensor data using multi-view stacking can improve classification performance in behavioral monitoring tasks [4]. Subsequent studies have combined multiple modalities including electrodermal activity, respiration rate, and skin temperature to improve classification performance. Deep learning approaches have shown strong results on physiological time-series data, and wearable-based stress datasets have enabled reproducible benchmarking of such methods [5].

Fuzzy logic has been applied to health monitoring systems due to its interpretability and ability to handle imprecise sensor measurements. Rule-based fuzzy inference systems have been used for pain assessment, fatigue detection, and mental workload estimation. Despite these advances, most existing datasets are small, short-duration, and collected under controlled conditions, limiting ecological validity. The SSAQS dataset [1] addresses this gap with longitudinal, ecologically valid data from a full academic semester.

III. DATASET AND FEATURES

The SSAQS dataset [1] was collected from 35 undergraduate volunteers at two Mexican universities during one academic semester (February–July 2025). Participants wore Fitbit Inspire 3 devices continuously and provided daily self-reports of stress and anxiety via a mobile application. The dataset achieves over 80% questionnaire compliance and contains validated Fitbit sensor measurements across five months of monitoring.

A. Data Collection

Seven data streams are recorded per student: HRV (5-minute RMSSD, low-frequency and high-frequency spectral power), blood oxygen saturation (SpO₂, recorded at minute intervals), step count, activity level classification (SEDENTARY, LIGHTLY_ACTIVE, FAIRLY_ACTIVE, VERY_ACTIVE), sleep quality (overall score and deep sleep duration), Fitbit's proprietary daily stress score, and daily questionnaire responses (0–100 stress and anxiety ratings). A total of 3,131 student-days are available after preprocessing.

B. Feature Engineering

Raw sensor streams are aggregated to daily summaries. HRV features include the daily mean and standard deviation of RMSSD and mean LF/HF spectral power. SpO₂ measurements below 70% are removed as Fitbit sensor artifacts. Physical activity is summarized as total daily step count, total active minutes, and the ratio of active minutes to total recorded minutes. Sleep is represented by the overall Fitbit sleep score and deep sleep duration in minutes. Table II summarizes all 12 input features.

TABLE II. INPUT FEATURE DESCRIPTION

Feature	Description
hrv_rmssd_mean	Mean daily RMSSD (HRV)
hrv_rmssd_std	Std. dev. of daily RMSSD
hrv_lf_mean	Mean low-frequency HRV power
hrv_hf_mean	Mean high-frequency HRV power
spo2_mean	Mean blood oxygen saturation
spo2_min	Minimum blood oxygen saturation
steps_total	Total daily step count
active_ratio	Ratio of active to total minutes
active_minutes	Total active minutes per day
sleep_score	Overall Fitbit sleep quality score
deep_sleep_min	Deep sleep duration (minutes)
fitbit_stress	Fitbit built-in daily stress score

C. Target Variable

Daily self-reported stress scores are discretized into three classes using per-student percentile thresholds: Low (below 33rd percentile), Medium (33rd–67th percentile), and High (above 67th percentile). Per-student thresholding accounts for individual differences in stress perception and reporting tendencies, resulting in balanced class distributions: Low = 1,110, Medium = 933, High = 1,088 samples.

IV. METHODOLOGY

A. Preprocessing

Missing sensor values are imputed using each student's own daily mean; remaining missing values are replaced with the global feature median across all students. All 12 features are scaled to [0, 1] using min-max normalization prior to model training to ensure comparability across sensors with different physical units and measurement scales.

B. Models

Five models are implemented and compared, representing different approaches to classification, from classical ensemble methods to deep learning and rule-based systems. This diversity allows a comprehensive evaluation of how well

each approach handles wearable sensor data that is small, noisy, and highly personal.

Random Forest was selected as a strong baseline. It constructs an ensemble of 100 decision trees, each trained on a bootstrap sample of the data, and combines their predictions by majority vote. Random Forest works well on small tabular datasets and is resistant to overfitting. It was also chosen because its built-in feature importance scores allow us to identify which sensor signals are most predictive of stress.

LightGBM was chosen because it is specifically designed for structured tabular data. It trains trees sequentially, where each new tree corrects the residual errors of the previous ensemble using second-order gradient information. Given that our dataset consists of 12 engineered daily features rather than raw signals, LightGBM was expected to be a strong performer.

The 1D CNN was selected because convolutional filters can detect local co-activation patterns across the 12 features. For example, the CNN can identify combinations where HRV and SpO₂ are simultaneously low. It applies two convolutional layers with 32 and 64 filters (kernel size 3, padding 1) separated by ReLU activations, followed by adaptive average pooling, 30% dropout, and a linear classification layer.

The LSTM was included because stress on a given day may be influenced by physiological patterns from previous days. A student who has had poor sleep or low activity over several consecutive days is more likely to report high stress. The LSTM uses a single recurrent layer with 64 hidden units to model these temporal dependencies. Both CNN and LSTM were trained for 15 epochs with batch size 64 and the Adam optimizer (learning rate 0.001) using cross-entropy loss.

Fuzzy Logic was chosen for its full interpretability. Rather than learning from data, it applies three weighted rules based on clinical knowledge: high HRV indicates lower stress (weight 0.4), high SpO₂ indicates lower stress (weight 0.3), and a high sleep score indicates lower stress (weight 0.3). The combined score is thresholded at 0.35 and 0.65 to produce Low, Medium, and High predictions. This model is the only one where every prediction can be explained in plain language, which is an important property in healthcare applications.

C. Validation Protocol

All models are evaluated using Leave-One-Student-Out (LOSO) cross-validation. In each of 35 folds, one student is held out as the test set while the model trains on the remaining 34 students. This protocol simulates deployment to a new individual who provided no training data and yields an unbiased estimate of generalization performance. Three metrics are reported: weighted accuracy, weighted F1-score (to account for class imbalance), and mean absolute error (MAE), which penalizes predictions proportionally to their distance from the true label.

V. RESULTS AND DISCUSSION

Table I summarizes classification performance across all five models under LOSO cross-validation. Fig. 1 provides a visual comparison of accuracy, F1-score, and MAE. Fig. 2

shows the LSTM confusion matrix. Fig. 3 shows feature importances from the Random Forest model.

TABLE I. CLASSIFICATION RESULTS (LOSO)

Model	Accuracy	F1-Score	MAE
Random Forest	0.3500	0.3439	0.9224
LightGBM	0.3552	0.3532	0.8809
CNN	0.3596	0.2559	0.9828
LSTM	0.3587	0.2900	0.9729
Fuzzy Logic	0.3012	0.2059	0.7400

LightGBM achieves the highest accuracy (0.3552) and weighted F1-score (0.3532), confirming that gradient boosting is well-suited for tabular physiological sensor data. CNN achieves comparable accuracy (0.3596) but a substantially lower F1-score (0.2559), suggesting it predicts the majority class (Low) more often than other classes. LSTM achieves similar accuracy (0.3587) with a moderate F1-score (0.2900). Random Forest performs competitively with accuracy 0.3500 and F1 0.3439.

Fuzzy Logic achieves the lowest accuracy (0.3012) and F1-score (0.2059), reflecting the limitations of a fixed rule-based system that cannot adapt to individual physiological baselines. However, it achieves the lowest MAE (0.7400), indicating that its predictions are closer to the true stress label on average, a clinically meaningful property for a monitoring system where the cost of a large prediction error exceeds that of a small one.

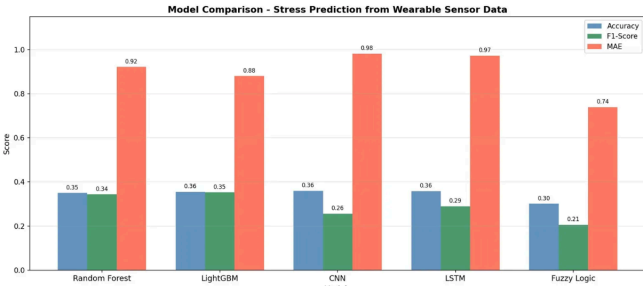


Fig. 1. Model comparison: Accuracy, F1-Score, and MAE across all five models under LOSO cross-validation.

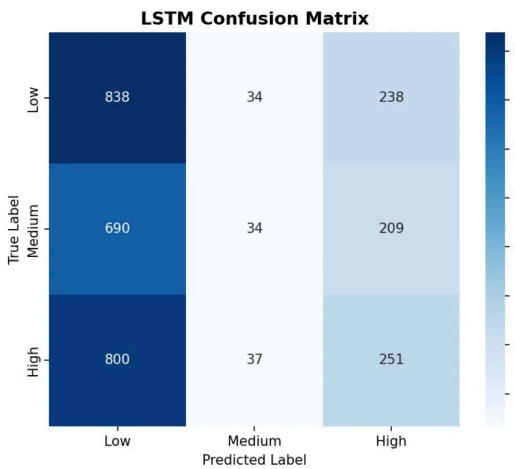


Fig. 2. LSTM confusion matrix. The model classifies Low stress most accurately (838 correct), while Medium stress is most frequently misclassified as Low (690 incorrect).

Fig. 2 reveals a systematic bias: all models show a strong tendency to predict Low stress. This is consistent with the known difficulty of cross-person stress classification under LOSO validation, where each test student has a unique physiological baseline that the model has never observed. The Medium stress class is the most challenging, with 690 of 933 samples misclassified as Low by the LSTM model.

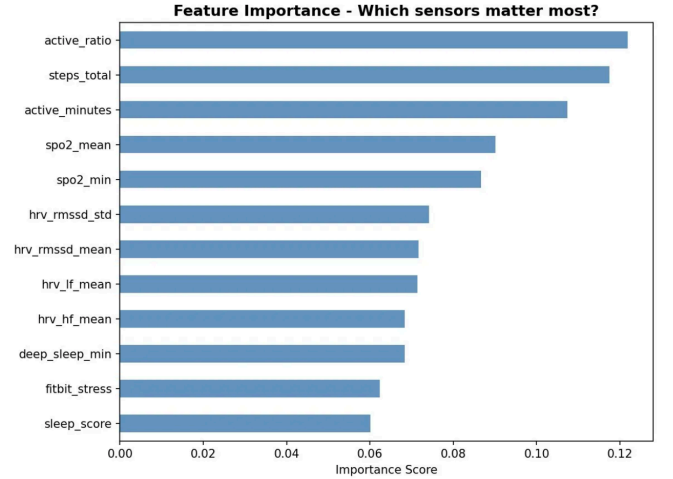


Fig. 3. Random Forest feature importance scores. Physical activity features (active_ratio, steps_total, active_minutes) are the strongest predictors of self-reported stress.

Fig. 3 shows that physical activity features (active_ratio, steps_total, active_minutes) are the three most important predictors, with importance scores of 0.122, 0.119, and 0.107 respectively. This finding is consistent with research linking sedentary behavior to elevated stress and poor mental health outcomes in student populations. Blood oxygen saturation (spo2_mean = 0.090) and HRV features rank in the middle tier. Sleep-related features and Fitbit's proprietary stress score rank lowest, possibly because they are derived metrics that aggregate information already captured by other features.

The overall accuracy values (~35%) reflect the fundamental challenge of cross-person stress prediction. Stress is a highly individualized phenomenon; a model trained on 34 students must generalize to a 35th student with a different physiological baseline, stress response pattern, and reporting style. This inter-individual variability is the primary bottleneck in wearable-based stress detection research and motivates personalized modeling approaches.

VI. CONCLUSION

This paper presented a systematic comparison of five machine learning approaches for student stress detection using Fitbit wearable sensor data from the SSAQS dataset. Under rigorous LOSO cross-validation, LightGBM achieved the best accuracy and F1-score, demonstrating the effectiveness of gradient boosting for tabular physiological data. Fuzzy Logic, despite lower classification accuracy, produced the smallest prediction errors (MAE = 0.7400), making it suitable for applications where the magnitude of error matters more than binary correctness.

Feature importance analysis revealed that physical activity measures are the strongest predictors of self-reported stress, a finding that has practical implications for intervention design. Students with low daily activity are more likely to report high stress, suggesting that activity-based interventions may be a viable complement to academic mental health support programs.

Future work should explore personalized modeling, where a global model is fine-tuned using a small number of labeled samples from the target student (few-shot adaptation). Multi-day temporal windows could be used as LSTM input sequences to better capture the longitudinal dynamics of academic stress. Incorporating contextual features such as exam schedules, assignment deadlines, and weather data may further improve predictive performance. Finally, the dataset could be augmented with speech or text data to add linguistic indicators of stress.

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<https://github.com/kubrasag/student-stress-detection-from-wearables>

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